

20251026 Offset Relation Derivation

Here we start with two continuity equations from Lilly+2013 or Wang+2021:

$$\begin{aligned}\frac{dM_g[t]}{dt} &= \Phi[t] - (1 - R)\text{SFR}[t] - \lambda\text{SFR}[t] \\ &= \Phi[t] - (1 - R + \lambda)\text{SFR}[t]\end{aligned}\quad (1)$$

Here $M_g[t]$ is the gas mass, $\Phi[t]$ is the inflow rate, R is the returned fraction from stars and λ is the mass loading factor. So these 3 terms are gas mass changes due to inflow, loss to star formation and loss to outflow.

$$\frac{dM_Z[t]}{dt} = y(1 - R)\text{SFR}[t] - Z[t](1 - R + \lambda)\text{SFR}[t] + Z_{\text{in}}\Phi[t] \quad (2)$$

Here $M_Z[t] = Z[t] \cdot M_g[t]$ is the metal mass, y is the metal yield per stellar generation and Z_{in} is the metallicity of the inflow gas. Again, these 3 terms are metal mass changes due to gain from metal production, loss to star formation and gain from inflow.

Now we can expand Eq. 2 by product rule to get

$$Z[t]\frac{dM_g[t]}{dt} + M_g[t]\frac{dZ[t]}{dt} = y(1 - R)\text{SFR}[t] - Z[t](1 - R + \lambda)\text{SFR}[t] + Z_{\text{in}}\Phi[t] \quad (3)$$

And substitute the Eq. 1 to equation above and simplify it to get $\frac{dZ[t]}{dt}$

$$\begin{aligned}Z[t](\Phi[t] - (1 - R + \lambda)\text{SFR}[t]) + M_g[t]\frac{dZ[t]}{dt} &= y(1 - R)\text{SFR}[t] - Z[t](1 - R + \lambda)\text{SFR}[t] + \\ &M_g[t]\frac{dZ[t]}{dt} = y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z[t])\Phi[t] \\ \frac{dZ[t]}{dt} &= \frac{1}{M_g[t]}(y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z[t])\Phi[t])\end{aligned}$$

In the equilibrium state, metallicity Z_{eq} will be independent on time t , then we can force Eq. 4 to be 0 and rearrange to get the expression of Z_{eq} in a function of $\text{SFR}[t]$ and $\Phi[t]$:

$$\begin{aligned}\frac{dZ[t]}{dt} &= \frac{1}{M_g[t]}(y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z_{\text{eq}})\Phi[t]) = 0 \\ y(1 - R)\text{SFR}[t] + (Z_{\text{in}} - Z_{\text{eq}})\Phi[t] &= 0 \\ Z_{\text{eq}} &= Z_{\text{in}} + y(1 - R)\frac{\text{SFR}[t]}{\Phi[t]}\end{aligned}\quad (5)$$

Since the inflow metallicity is much less than equilibrium metallicity ($Z_{\text{in}} \ll Z_{\text{eq}}$), we can approximate the equation above as

$$Z_{\text{eq}} \approx y(1 - R)\frac{\text{SFR}[t]}{\Phi[t]} \quad (6)$$

Now let consider a small perturbation of Z_{eq} (offset from the mean value). Since y and R are all constant here, we have

$$\delta Z_{\text{eq}} = y(1 - R)\delta\left(\frac{\text{SFR}[t]}{\Phi[t]}\right) \quad (7)$$

Because we are interested in the offset in $12 + \log_{10}(\text{O}/\text{H})$, we have

$$\begin{aligned} \delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}}) &= \delta \log_{10}(Z_{\text{eq}}) \\ &= \frac{1}{\ln 10} \frac{\delta Z_{\text{eq}}}{Z_{\text{eq}}} \\ &= \frac{1}{\ln 10} \frac{y(1 - R)\delta\left(\frac{\text{SFR}[t]}{\Phi[t]}\right)}{y(1 - R)\frac{\text{SFR}[t]}{\Phi[t]}} \\ &= \frac{1}{\ln 10} \frac{\delta(\text{SFR}[t]/\Phi[t])}{\text{SFR}[t]/\Phi[t]} \\ &= \frac{1}{\ln 10} \frac{\frac{\delta(\text{SFR}[t]/\Phi[t])}{\delta \text{SFR}[t]} \delta \text{SFR}[t] + \frac{\delta(\text{SFR}[t]/\Phi[t])}{\delta \Phi[t]} \delta \Phi[t]}{\text{SFR}[t]/\Phi[t]} \quad (8) \\ &= \frac{1}{\ln 10} \frac{\frac{1}{\Phi[t]} \delta \text{SFR}[t] - \frac{\text{SFR}[t]}{\Phi^2[t]} \delta \Phi[t]}{\text{SFR}[t]/\Phi[t]} \\ &= \frac{1}{\ln 10} \left(\frac{\delta \text{SFR}[t]}{\text{SFR}[t]} - \frac{\delta \Phi[t]}{\Phi[t]} \right) \\ &= \frac{1}{\ln 10} (\delta \ln(\text{SFR}[t]) - \delta \ln(\Phi[t])) \\ &= \delta \log_{10}(\text{SFR}[t]) - \delta \log_{10}(\Phi[t]) \end{aligned}$$

Thus, we can see that the sign of $\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}})$ is only determined by relative change of $\log_{10}(\text{SFR}[t])$ and $\log_{10}(\Phi[t])$.

More specific, by $\text{SFR}[t] = \epsilon[t]M_g[t]$, where $\epsilon[t]$ is the star formation efficiency, we have

$$\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}}) = \delta \log_{10}(\epsilon[t]) + \delta \log_{10}(M_g[t]) - \delta \log_{10}(\Phi[t]) \quad (9)$$

Therefore, this leaves us 3 cases.

1. In the case that $\delta \log_{10}(\epsilon[t])$ dominants, if $\delta \log_{10}(\epsilon[t])$ increases, both $\delta(12 + \log_{10}(\text{O}/\text{H})_{\text{eq}})$ and $\delta \log_{10}(\text{SFR}[t])$ increase immediately, so we can observe the positive correlation between them.
2. However, if $\delta \log_{10}(\Phi[t])$ dominates, even though $\delta \log_{10}(\text{SFR}[t])$ is still positive, due to the minus sign and relatively large change in $\delta \log_{10}(\Phi[t])$, we can observe negative correlation.
3. If the change of $\log_{10}(\text{SFR}[t])$ and $\log_{10}(\Phi[t])$ is comparable, that lead to nearly stationary $12 + \log_{10}(\text{O}/\text{H})$, which show no correlation. In this scenario, it requires $\Phi[t]$ is linearly proportional to $\text{SFR}[t]$.